

Rectangular patch antenna

Rectangular patch antenna (patch-antenna)

Probe-fed microstrip patch on a finite RO4003C-like substrate and ground plane.

Reference: openEMS tutorial 'Simple Patch Antenna' (T. Liebig)

Simulated	2026-09-24T23:38:20+0300
Report generated	2026-09-24 23:38
Generator	Python 3.13.7
Solver	openEMS 0.37.0rc3.post1.dev11+g12cd91de2 — FDTD (Yee, staircase)
CSXCAD	0.7.0rc3.post1.dev1+gbd2c13339
Host	Darwin arm64, Apple M5 Pro

Key results

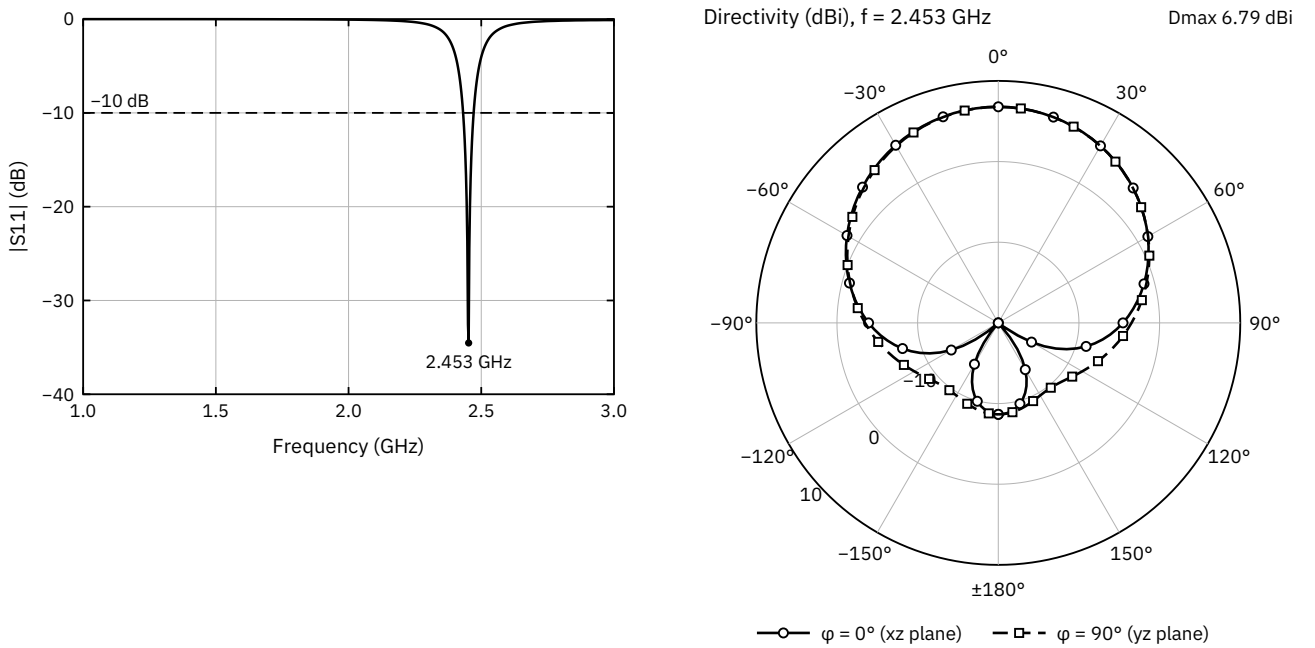
✓ Converged: field energy -79.3 dB at timestep 18,468 (≤ -60 dB criterion), 16.1 s solver time.

Matched bands ($|S_{11}| < -10$ dB)

Center (GHz)	Best match (GHz)	Range (GHz)	Bandwidth (MHz)	Fractional BW (%)	$ S_{11} $ min (dB)
2.451	2.453	2.433 – 2.470	38	1.5	-34.53

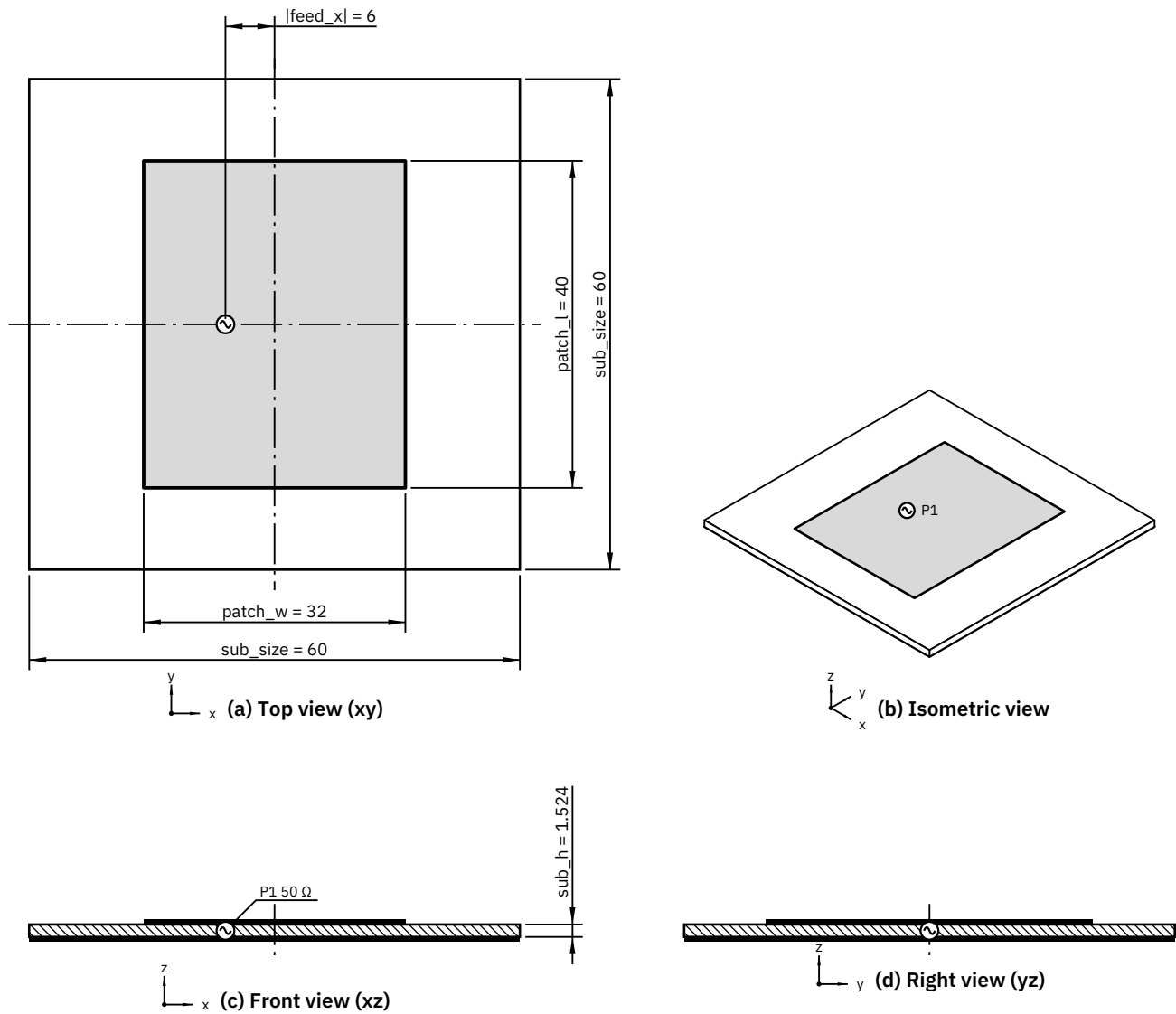
Far field

f (GHz)	Dmax (dBi)	Gain (dBi)	Realized gain (dBi)	Rad. efficiency (%)
2.453	6.79	6.63	6.63	96.5



At a glance: $|S_{11}|$ versus frequency and directivity cuts at 2.453 GHz (details on the following pages).

Technical drawing



NOTES

1. All dimensions in mm.
2. Ground plane = Substrate outline (60 × 60).
3. Metal parts are PEC; zero-thickness sheets are drawn with a minimum line width (solid black when seen edge-on).
4. P1: lumped port, 50 Ω , along z, gap 1.524 mm, excited.

Figure 1. Rectangular patch antenna: orthographic views (third-angle projection) and isometric view; dimensions in mm, labelled with the model parameter where one matches.

Parameters

Key	Label	Value	Default	Unit
patch_w	Patch length (resonant, x)	32	32	mm
patch_l	Patch width (y)	40	40	mm
sub_size	Substrate / ground size	60	60	mm
sub_h	Substrate thickness	1.524	1.524	mm
eps_r	Substrate permittivity	3.38	3.38	—
tan_d	Loss tangent	0.001	0.001	—
feed_x	Feed position (x)	−6	−6	mm
mesh_div	Mesh: cells per λ at f max	30	30	—
sub_cells	Mesh cells across the substrate thickness	4	4	—
f_min	Band start	1	1	GHz
f_max	Band stop	3	3	GHz

Geometry and materials

Part	Type	Primitives	ϵ_r	$\tan \delta$	Size (mm)
Patch	PEC	1	—	—	32 × 40 × 0
Substrate	Dielectric	1	3.38	0.001 @ 2.000 GHz	60 × 60 × 1.524
Ground plane	PEC	1	—	—	60 × 60 × 0

Port	Type	R (Ω)	Direction	Start (mm)	Stop (mm)	Excited
1	lumped	50	z	−6, 0, 0	−6, 0, 1.524	yes

Solver, mesh and run

Method	FDTD (Yee, staircase)
Excitation	gaussian-derivative (DC-free), 1.000 – 3.000 GHz
Boundaries	x- MUR, x+ MUR, y- MUR, y+ MUR, z- MUR, z+ MUR
End criterion	−60 dB, at most 30000 timesteps
Mesh	68 × 68 × 57 = 263,568 cells, cell size 0.3810 – 3.263 mm
Domain	(−100, −100, −50) to (100, 100, 100) mm
Converged	yes, final energy −79.3 dB at timestep 18,468
Timesteps	18468
Solver time	16.06 s (wall 16.32 s), 317.5 MCells/s, CPU, threads 4

Input reflection coefficient

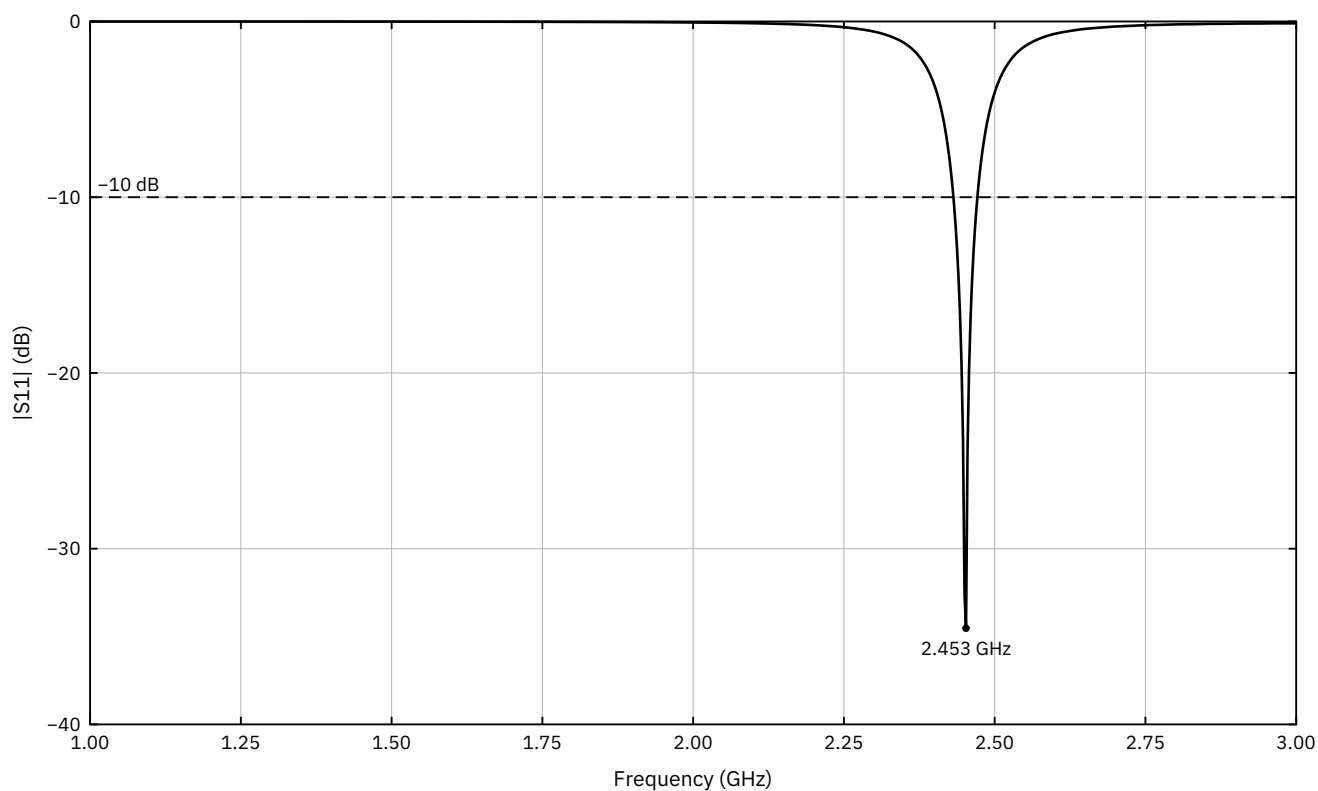


Figure 2. $|S_{11}|$ versus frequency; the dashed line marks -10 dB and the dots the best-match frequencies.

Band	f_lo (GHz)	f_hi (GHz)	Center (GHz)	Best match (GHz)	S11 min (dB)
1	2.433	2.470	2.451	2.453	-34.53

Input impedance

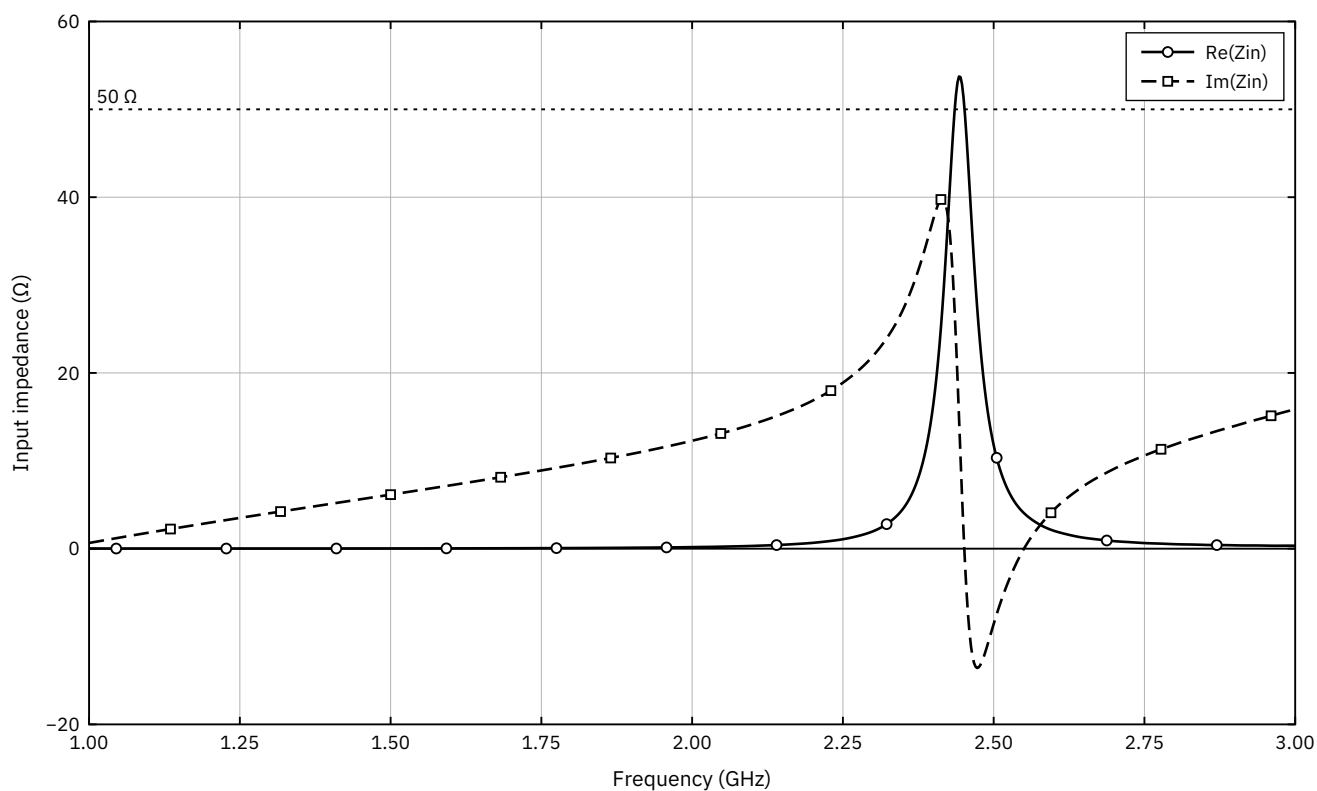


Figure 3. Real (solid, circles) and imaginary (dashed, squares) part of Z_{in} ; the dotted line marks the port reference impedance.

Smith chart

Input reflection coefficient, $Z_0 = 50 \, \Omega$

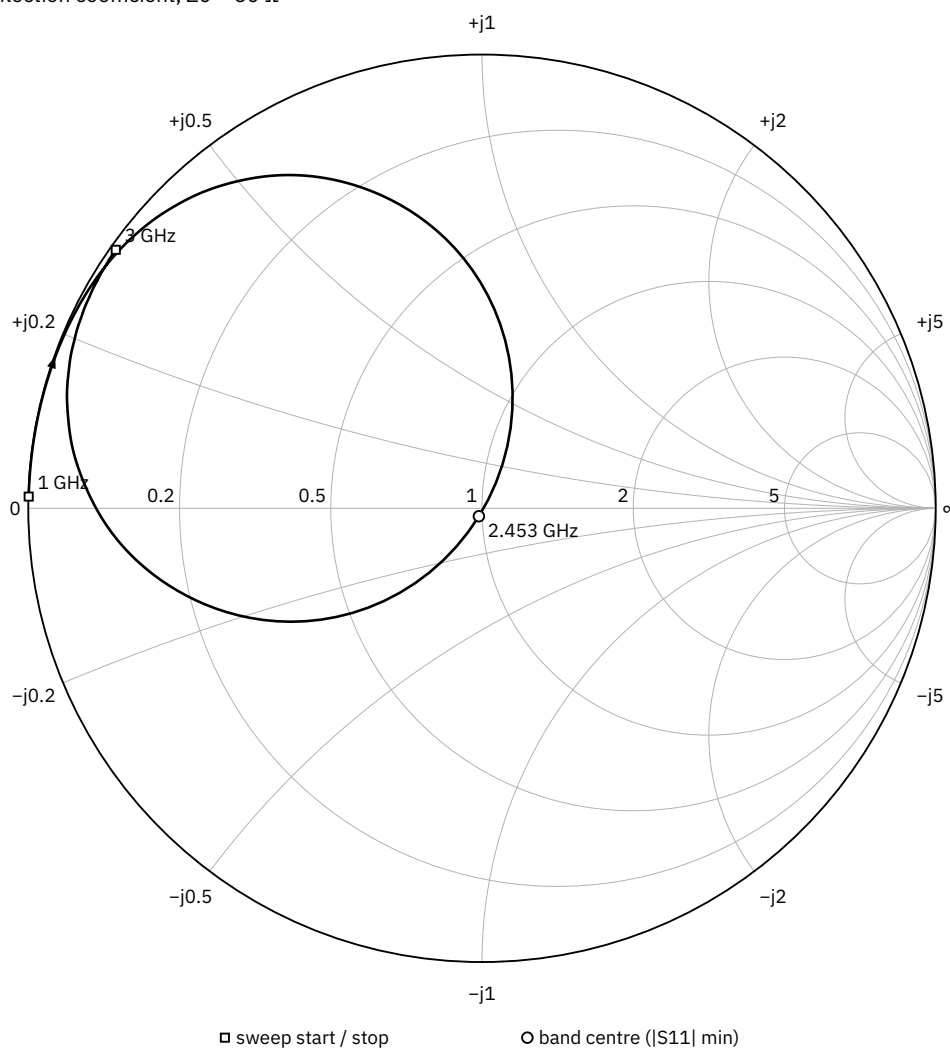


Figure 4. S_{11} on the Smith chart (normalised to the port impedance) from 1.000 to 3.000 GHz.

Radiation pattern at 2.453 GHz

Directivity (dBi), f = 2.453 GHz

Dmax 6.79 dBi

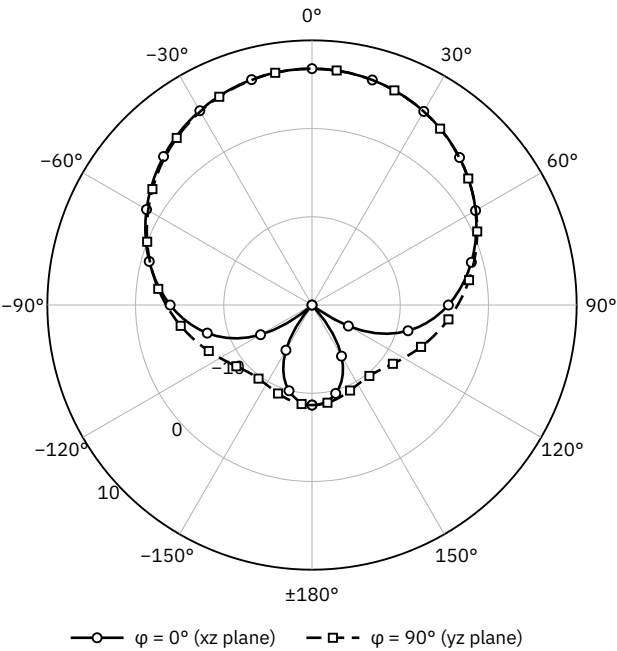


Figure 5. Directivity cuts $\phi = 0^\circ$ (xz plane, solid) and $\phi = 90^\circ$ (yz plane, dashed); θ measured from +z.

Dmax	6.79 dBi
Gain / realized gain	6.63 dBi / 6.63 dBi
Radiation efficiency	96.5 %
Mirror planes	0

Reproduce

Run the same model with the fairbeam command (openEMS installed, see the Fairbeam README):

```
fairbeam run '<workspace>/models/patch_antenna.py' --fields --threads 4
```

Runs the model file in the models folder of your Fairbeam workspace (<workspace>); pass the path of the model or design file if it is named differently than patch_antenna.py. Only parameters that differ from the defaults are listed; project.json holds the full setup.

Files and formats

project.json	The complete Fairbeam bundle (schema fairbeam.project/1): geometry, mesh, solver setup, run statistics and all results.
data/s11.s1p	Touchstone v1, option line “# GHz S RI R <port impedance>”: frequency in GHz, Re and Im of S11. Most RF tools read it directly, so it can be overlaid on other S11 results.
data/*.csv	Sweep (f, S11 dB, S11, VSWR, Zin), matched bands, far-field summary, directivity over θ and φ (long format) and port time signals.
drawings/, figures/	Black-and-white vector drawings and publication figures (SVG; the A3 drawing also as PDF).
cst/*.bas	CST-compatible VBA macro that rebuilds the model.
README.md	This report as text, with the list of every file.

Notes

openEMS uses a staircase (Yee) FDTD mesh: slanted edges snap to mesh lines and zero-thickness sheets sit on grid planes. Expect resonances to differ by a few percent from conformal solvers unless the mesh is fine on metal edges; check mesh convergence before trusting a number.